# CHAPTER 01

# INTRODUCTION

## 1. Introduction

Oral cancer is one of the most prevalent and life-threatening cancers worldwide, particularly in developing regions where late diagnosis, limited access to specialists, and inefficient healthcare workflows significantly affect patient survival rates. Despite advancements in medical science, early detection of oral cancer remains a major challenge due to dependence on manual clinical examination, subjective assessment of symptoms, and the lack of integrated digital systems for screening, diagnosis, and treatment management. In many healthcare institutions, patient records, biopsy tracking, and oncology care are still managed using paper-based or disconnected systems, leading to delays, data loss, poor coordination among specialists, and reduced quality of care.

Recent developments in Machine Learning (ML) and digital health technologies have demonstrated strong potential in improving disease prediction, clinical decision support, and patient management. However, many existing oral cancer detection systems rely heavily on image-based analysis or expensive diagnostic tools, which may not always be accessible in resource-limited settings. There is a growing need for an intelligent, cost-effective, and clinically practical solution that leverages routinely collected clinical parameters while also providing end-to-end patient care management.

OraTrack is designed to address these challenges by introducing an ML-powered oral cancer detection and smart oncology patient management system. The platform integrates intelligent risk prediction with automated biopsy workflows and structured oncology care within a single digital ecosystem. By analyzing clinically relevant inputs such as lesion characteristics, symptom duration, lifestyle factors, and patient medical history provided by healthcare professionals and patients, OraTrack facilitates early risk assessment and stratification. Beyond detection, the system ensures seamless coordination between doctors, lab technicians, and oncologists through automated task assignment, QR-based biopsy tracking, real-time notifications, and centralized patient records.Furthermore, OraTrack extends its functionality beyond diagnosis to continuous oncology care, enabling treatment planning, symptom monitoring, patient engagement, and emergency support. By bridging the gap between screening, diagnosis, and long-term cancer management, OraTrack aims to reduce diagnostic delays, improve clinical efficiency, enhance patient–doctor communication, and ultimately contribute to better treatment outcomes in oral cancer care.

## 2. Problem Statement

Oral cancer poses a serious healthcare challenge due to high mortality rates primarily caused by late diagnosis, inefficient screening processes, and poor coordination among healthcare providers. In many hospitals and diagnostic centers, oral cancer screening relies on manual clinical evaluation and subjective judgment, which increases the risk of missed or delayed detection. The absence of intelligent decision-support systems makes it difficult for doctors to accurately assess cancer risk during early stages, especially in resource-constrained settings.

Additionally, biopsy procedures and laboratory workflows are often managed using paper-based or loosely connected systems, resulting in delays, lack of transparency, and miscommunication between doctors and lab technicians. Tracking biopsy status, report generation, and turnaround time becomes difficult, directly impacting treatment initiation. Once cancer is confirmed, patient information is frequently fragmented across departments, forcing oncologists to reassess cases from scratch and leading to further delays in treatment planning.Moreover, oncology patient management lacks continuous monitoring and structured follow-up mechanisms. Patients undergoing treatment often have limited means to report daily symptoms, therapy side effects, or emergency conditions in real time. The absence of centralized digital communication and alert systems reduces timely clinical intervention and compromises patient safety.

## 3. Scope and Relevance of the Project

The scope of the ML Powered Oral Cancer Detection and Smart Oncology Patient Management System (OraTrack) covers the entire continuum of oral cancer care, from early risk assessment to long-term oncology management. The system enables early oral cancer screening using machine learning based on doctor and patient entered clinical parameters such as lesion size, ulcer duration, pain intensity, bleeding, lifestyle habits, and medical history, making it suitable even for resource-limited healthcare settings. For patients identified with moderate or high risk, OraTrack automatically initiates biopsy requests and ensures transparent tracking of the biopsy process through QR-based status updates and digital laboratory workflows.

Beyond diagnosis, the platform provides seamless coordination between doctors, lab technicians, and oncologists by maintaining centralized patient records and real-time notifications. Once cancer is confirmed, the system supports structured oncology care, including treatment planning, therapy scheduling, progress evaluation, and continuous follow-up. Patients can actively participate in their care by reporting daily symptoms, treatment side effects, and medication updates, while emergency alert features enable timely medical intervention. Overall, OraTrack is highly relevant as it improves early detection, reduces diagnostic and workflow delays, enhances doctor–patient communication, and aligns with modern healthcare digitization efforts, ultimately contributing to better oral cancer treatment outcomes.

## 4. Objectives

- To develop an ML-powered system for early detection and risk prediction of oral cancer using doctor and patient entered clinical parameters.

- To classify patients into Normal, Low, Moderate, and High-risk categories to support accurate clinical decision-making.

- To automate biopsy request generation for moderate- and high-risk cases and streamline the biopsy workflow.

- To implement QR-based biopsy tracking to ensure transparency and reduce delays in laboratory processes.

- To enable lab technicians to update biopsy status, upload digital pathology reports, and record processing time efficiently.

- To ensure seamless transfer of confirmed cancer cases to oncologists with complete clinical history and reports.

- To provide a structured oncology patient management module for treatment planning, therapy scheduling, and progress monitoring.

- To facilitate continuous patient engagement through daily symptom reporting, medication updates, and treatment reminders.

- To support real-time doctor–patient communication and emergency alert mechanisms for critical situations.

- To create a centralized digital health record system that improves coordination, efficiency, and quality of oral cancer care.